low cost converyor based micro logistics system for warehouse

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Project Abstract

The proposed system is a low-cost conveyor-based micro logistics system designed to automate the movement and handling of small packages inside warehouses. The system uses a conveyor mechanism, sensors, a microcontroller, and an automatic sorting mechanism to reduce manual material handling and improve warehouse efficiency.

Problem Statement

- Manual movement of small packages is time-consuming and labor-intensive.
- Conventional warehouse automation systems are expensive for small-scale applications.

Need for the Project

- To develop an affordable warehouse automation solution.
- To reduce manual material handling and improve productivity.

Project Objectives

- Design and fabricate a compact conveyor-based logistics system.
- Automatically detect and transport packages.

Expected Learning Outcomes

- Learn conveyor mechanism and warehouse material-handling principles.
- Develop skills in sensor integration and microcontroller programming.

System Requirements

Hardware

- Conveyor belt and rollers, DC geared motorors, Motor driver, IR sensors, power supply.

Software

- Arduino IDE and embedded C/C++.

Methodology / Workflow

Package Input → Package Detection → Identification → Conveyor Movement → Destination Detection → Automatic Sorting → Storage/Dispatch

System Architecture

- Sensors detect and identify the package and send information to the microcontroller.
- The controller operates the conveyor motor and sorting actuator according to the package destination.

Block Diagram

Package Input → IR/RFID Sensors → Microcontroller → Motor Driver → Conveyor → Servo Sorting Mechanism → Storage/Dispatch Bin.

Project Development Process

Problem Identification → Concept Design → Mechanical Design → Component Selection → Fabrication → Programming → Integration → Testing → Evaluation .

Implementation

- The conveyor frame, rollers, belt, motor, sensors, controller, and sorting mechanism are integrated to create a working warehouse micro-logistics prototype.

Algorithms / Models Used

- Rule-based decision-making is used for package sorting.
- The controller selects the appropriate destination based on sensor input.

QR Code — Project Report (PDF) | Source Code Repository | Demonstration Video

Dataset / Input Data (if applicable)

Different packages with varying sizes, weights, and destination categories are used for testing the conveyor-based transportation and sorting system.

Experimental Setup

A small conveyor system with a motor, sensors, and microcontroller is set up to transport and sort different packages.

Results

The system automatically detects, transports, and sorts packages into designated warehouse zones. It reduces manual handling and improves material movement efficiency.

Performance Evaluation

Performance is evaluated using package detection accuracy, sorting accuracy, processing time, conveyor speed, and system reliability.

Testing and Validation

Different packages are tested repeatedly to verify accurate detection, smooth transportation, and proper sorting.

Challenges Encountered

- Different package sizes and weights can affect sensor detection.
- Conveyor belt alignment and servo positioning can affect sorting accuracy.

Solutions Implemented

- Sensors are calibrated and properly positioned for reliable package detection.
- Conveyor alignment and servo positioning are adjusted through repeated testing.

Innovation / NovelFeatures

- Automated package detection, transportation, and sorting with minimal human intervention.
- The system can be extended with IoT and AI technologies.

Applications

- Small warehouses, e-commerce warehouses, manufacturing industries, inventory areas, packing sections, and dispatch centers.

Future Scope

IoT-based monitoring and RFID tracking can be integrated. Future versions can include AI-based package classification, smart inventory monitoring, and automatic alerts.ing and automatic alerts.

Project Outcomes

Developed a low-cost conveyor-based logistics prototype that reduces manual handling and improves package transportation and sorting efficiency.

References

1. M. P. Groover, Automation, Production Systems, and Computer-Integrated Manufacturing, Pearson Education, 4th Edition, 2015, pp. 1-30.
2. R. K. Jain, Production Technology, Khanna Publishers, 17th Edition, 2016, pp. 450-480.

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